AP Calculus II
Notes 9.1
Sequences

A sequence is any function whose domain is the set of positive integers. However, it is common to represent sequences in subscript notation, as shown below:

$2\ ,\ 4\ ,\ 6\ ,\ 8\ ,\ \ldots\ ,\ n\ ,\ \ldots$ the entire sequence is denoted as:

Writing out the Terms of a Sequence

Ex. 1: Write out the first five terms of the following sequences.

a) $\{a_n\} = \{3 + (-1)^n\}$

b) $\{b_n\} = \left\{\frac{n}{1+3n}\right\}$

c) $\{e_n\} = \left\{\frac{(-1)^{n+1}}{n!}\right\}$

d) $\{c_n\} = \left\{\frac{n^2}{2^n-1}\right\}$

Limit of a Sequence

One major purpose of sequences is to see if the terms approach a certain value as the terms go on forever. These sequences are said to **converge**. For example:

The sequence $\left\{\frac{1}{2^n}\right\} =$ converges to

Definition of the Limit of a Sequence

Let L be a real number. The limit of a sequence $\{a_n\}$ is L, written as $\lim_{n\to\infty} a_n = L$, and if this limit exists, then the sequence **converges** to L. If the limit of a sequence does not exist, then the sequence **diverges**.

Additionally, let L be a real number. Let f be a function of a real variable such that $\lim_{x\to\infty} f(x) = L$. If $\{a_n\}$ is a sequence such that $f(n) = a_n$ for every positive integer n, then $\lim_{n\to\infty} a_n = L$.

Determining Convergence or Divergence

Ex. 2: Determine whether each of the following sequences converge or diverge.

a) $\{a_n\} = \{3 + (-1)^n\}$

b) $\{b_n\} = \left\{\frac{n}{1+3n}\right\}$

c) $\left\{\frac{3^n+\cos n\pi}{2n^2}\right\}$

d) $\left\{\frac{n^2}{e^{n-1}+1}\right\}$

e) $\left\{\cos\left(\frac{3}{n}\right)\right\}$

f) $\left\{\frac{(-1)^n n!}{(n+2)!}\right\}$

Pattern Recognition for Sequences

Ex. 3: Find the *n*th term of the sequences described below.

a) $4, \frac{8}{3}, \frac{16}{9}, \frac{32}{27}, \frac{64}{81}, \ldots$

b) $-\frac{3}{1}, \frac{9}{2}, -\frac{27}{6}, \frac{81}{24}, -\frac{243}{120}, \ldots$

c) $-1, 2, 7, 14, 23, \ldots$

d) $\frac{2}{1}, \frac{4}{3}, \frac{8}{5}, \frac{16}{7}, \frac{32}{9}, \ldots$

e) $\frac{(x-2)}{6}, \frac{(x-2)^2}{12}, \frac{(x-2)^3}{24}, \frac{(x-2)^4}{48} \ldots$

f) $1, -x, \frac{x^2}{2}, \frac{-x^3}{6}, \frac{x^4}{24}, \ldots$

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Notes 9.2
Series and Convergence

One important application of infinite sequences is the representation of infinite summations. The **<u>sum</u>** of an infinite <u>sequence</u> is an **infinite <u>series</u>**.

$$a_1 + a_2 + a_3 + \ldots + a_n + \ldots = \sum_{n=1}^{\infty} a_n$$

To find the sum of an infinite series (the limit), we examine the sequence of partial sums:

Consider the series: $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \ldots = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$

The sum of the first term is called the first <u>partial sum</u>. The second partial sum is the sum of the first 2 terms. The *n*th partial sum is the sum of the first *n* terms.

$\sum_{n=1}^{1} a_n = \frac{1}{2}$ $\qquad \sum_{n=1}^{2} a_n =$ $\qquad \sum_{n=1}^{3} a_n =$

$\sum_{n=1}^{4} a_n =$ $\qquad \sum_{n=1}^{5} a_n =$

What is $\sum_{n=1}^{x} a_n = ?$ $\qquad$ What is $\lim_{x \to \infty} \sum_{n=1}^{x} a_n$?

<u>Definitions of Convergent and Divergent Series</u>
If the sequence of partial sums $\{S_n\}$ converges to S, the infinite series $\sum_{n=1}^{\infty} a_n$ converges and has a sum S. If $\{S_n\}$ diverges, the series diverges.

Ex. 1: Determine if the <u>sequence</u> a_n converges and then determine if the <u>series</u> $\sum_{n=1}^{\infty} a_n$ converges.

a) $\sum_{n=1}^{\infty} \frac{1}{2}$

b) $\sum_{n=0}^{\infty} 3(0.1)^n$

Geometric Series:

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ar^3 + \cdots + ar^n + \cdots \qquad a \neq 0 \qquad \text{where } r \text{ is known as the}$$

Recall from Pre-Calculus that an infinite geometric series converges if $|r| < 1$ and the sum of this convergent series is $S_n = \frac{a_1}{1-r}$ where a_1 is the first term. The series diverges if $|r| \geq 1$.

Ex. 2: Verify the convergence of the series from the previous example using the Geometric Series Test.

a) $\sum_{n=1}^{\infty} \frac{1}{2}$

b) $\sum_{n=0}^{\infty} 3(0.1)^n$

c) $\sum_{n=0}^{\infty} \frac{8}{5^n}$

Ex. 3: Write the follow series in sigma notation. Then, determine whether the geometric series converges or diverges. If it converges, find the sum.

a) $\frac{4}{3}+\frac{4}{9}+\frac{4}{27}+\frac{4}{81}+\ldots$

b) $1-\frac{e}{3}+\frac{e^2}{9}-\frac{e^3}{27}+\frac{e^4}{81}+\ldots$

c) $\frac{4}{3}+\frac{16}{9}+\frac{64}{27}+\frac{256}{81}+\ldots$

Ex. 4: Consider the geometric series $\sum_{n=1}^{\infty} a_n$, where $a_n > 0$, and $\sum_{n=1}^{\infty} b_n$ for all n. The first term of both of the series is $a_1 = b_1 = 720$. The third term is $a_3 = 80$ and the fourth term is $b_4 = -90$. Determine the convergence of both of the series, if they do.

Properties of Infinite Series

Let $\sum^{\infty} a_n = A$, $\sum^{\infty} b_n = B$, and c is a real number, then

1. $\sum^{\infty} ca_n = cA$

2. $\sum^{\infty} (a_n \pm b_n) = A \pm B$

Ex. 5: Find $\sum_{n=1}^{\infty} \left[3(2)^{-2n} + \frac{2}{5^{n-1}} + \frac{(-4)^{n+1}}{5^n} \right]$

Ex. 6: Find the values of x for which the following series converges. For these values, write the sum of the series as a function of x.

a) $\sum_{n=1}^{\infty} \frac{x^n}{3^n}$

b) $\sum_{n=0}^{\infty} 4\left(\frac{x-3}{4}\right)^n$

<u>Nth Term Test for Divergence:</u>

As a review, find the limit of each **<u>sequence</u>** and then determine the convergence of the following **<u>series</u>**:

$$\sum_{n=1}^{\infty} 2 \qquad\qquad \sum_{n=1}^{\infty} 2\left(\frac{4}{3}\right)^n \qquad\qquad \sum_{n=1}^{\infty} \frac{3n}{5n+4}$$

This is a small insight of the **nth Term Test for Divergence**, which states that:

For a series $\sum_{n=1}^{\infty} a_n$, if $\lim_{n\to\infty} a_n \neq 0$, then the series diverges.

<u>What this does and does not mean:</u>

<u>DOES MEAN</u>	**<u>DOES NOT MEAN</u>**

<u>THIS TEST CANNOT DETERMINE CONVERGENCE!!!!!</u>

Ex. 7: Use the nth term test for divergence, and any others learned, to determine if the series diverges:

a) $\displaystyle\sum_{n=1}^{\infty}\left(\frac{6}{5}\right)^n$

b) $\displaystyle\sum_{n=1}^{\infty}\frac{6}{5^n}$

c) $\ln(1) + \ln\left(\frac{1}{2}\right) + \ln\left(\frac{1}{3}\right) + \ln\left(\frac{1}{4}\right) + \ldots$

d) $\displaystyle\sum_{k=1}^{\infty}\frac{6^k}{k^2+k+1}$

e) $\displaystyle\sum_{n=1}^{\infty}\frac{5n^2}{n^2+5n+6}$

f) $1 - \frac{2}{3} + \frac{4}{9} - \frac{8}{27} + \ldots$

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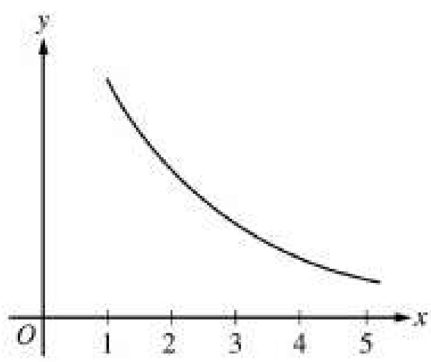
Notes 9.3
Integral Test and p-Series

Over the next two sections, we will learn several convergence tests that apply to series with *positive* terms.

The Integral Test

If $f(x)$ is positive, continuous, and decreasing for $x \geq 1$ and $a_n = f(n)$, then:

Ex. 1:



Graph of g

For $x \geq 1$, the continuous function g is decreasing and positive. A portion of the graph of g is shown above. For $n \geq 1$, the nth term of the series $\sum_{n=1}^{\infty} a_n$ is defined by $a_n = g(n)$. If $\int_1^{\infty} g(x)\,dx$ converges to 8, which of the following could be true?

(A) $\sum_{n=1}^{\infty} a_n = 6$

(B) $\sum_{n=1}^{\infty} a_n = 8$

(C) $\sum_{n=1}^{\infty} a_n = 10$

(D) $\sum_{n=1}^{\infty} a_n$ diverges

Ex. 2: Determine the convergence of the following series:

a) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{n^2+1}$

b) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2+1}$

p-Series and Harmonic Series

The second test for convergence of a series has a simple arithmetic form:

$$\sum_{n=1}^{\infty} \frac{1}{n^p} =$$

which is known as a ***p*-series**, where p is a positive constant. For $p = 1$, the series:

$$\sum_{n=1}^{\infty} \frac{1}{n} =$$

is known as the **harmonic series**. The Greek used this in music to create the length of strings of the same material, diameter, etc… that produce harmonic tones.

The Integral Test is useful in determining the convergence and divergence of this p −series.

Ex. 3: Apply the Integral Test to determine the p −series rules:

a) $\displaystyle\sum_{n=1}^{\infty}\frac{8}{\sqrt{n}}$

b) $5+\frac{5}{4}+\frac{5}{9}+\frac{5}{16}+\cdots$

Convergence of *p*-Series proof:

$$\sum_{n=1}^{\infty}\frac{1}{n^p}=$$

a) converges if

b) diverges if

Note that for a p −series, the p −value is considered positive when in the denominator and could possibly have a constant multiple. This may require a little rewriting or simplification to get a series in this form.

Ex. 4: Determine if the following series converge or diverge:

a) $\displaystyle\sum_{k=1}^{\infty} \frac{4}{k^{1.001}}$

b) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{3}$

c) $1 + \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{3}} + \frac{1}{\sqrt[3]{4}} + \ldots$

d) $\frac{4}{205} + \frac{5}{207} + \frac{6}{209} + \frac{7}{211} + \frac{8}{213} + \ldots$

e) $\frac{6}{2} + \frac{6}{5} + \frac{6}{10} + \frac{6}{17} + \frac{6}{26} + \ldots$

f) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^{n-1}$

AP Calculus II
Notes 9.4
Comparisons of Series

For all the convergence tests that we have used so far, the series have been very simple and have fit into each of the tests exactly. A small deviation from these special characteristics can make a test inapplicable. These tests are very similar to integration rules. For example, check out the following pairs:

a) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{2^n}$ is

b) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3}$ is

c) $\displaystyle\sum_{n=1}^{\infty} \frac{n}{(n^2+1)^2}$ is

$\displaystyle\sum_{n=0}^{\infty} \frac{n}{2^n}$ is

$\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^3+1}$ is

$\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{(n^2+1)^2}$ is

In this section we will learn two additional tests for positive term series. These two tests will allow us to determine the convergence or divergence of a whole new set of series. They will allow us to **compare** a series having complicated terms with a **simpler** series whose convergence or divergence is known.

Visual Representation of Comparison Tests

The Direct Comparison Test

Let $0 \le a_n \le b_n$ for all n.

1) If $\displaystyle\sum_{n=1}^{\infty} b_n$ converges,

2) If $\displaystyle\sum_{n=1}^{\infty} b_n$ diverges,

3) If $\displaystyle\sum_{n=1}^{\infty} a_n$ diverges,

4) If $\displaystyle\sum_{n=1}^{\infty} a_n$ converges,

Ex. 1: Determine the convergence or divergence of the following series:

a) $\displaystyle\sum_{n=0}^{\infty} \frac{1}{2+3^n}$

b) $\displaystyle\sum_{k=0}^{\infty} \frac{3}{4+\sqrt{k}}$

c) $\displaystyle\sum_{k=2}^{\infty} \frac{\ln k}{k^4+1}$

Sometimes, we can see that a series resembles a ***p*-series** or a **geometric series**, but we can't establish the necessary comparison that satisfies the conditions of the Direct Comparison Test. When this happens, you may be able to apply the second comparison test known as the **Limit Comparison Test**.

Limit Comparison Test

Suppose that $a_n > 0, b_n > 0$, and $\lim_{n \to \infty} \left(\frac{a_n}{b_n}\right) = L$ where L is **finite** and **positive**. This ratio will say that as the terms go on to infinity, they are off by a factor of a constant multiple. Therefore, the sum of these terms will be off by that constant multiple (about). Then, the two series either both converge or both diverge.

Ex. 2: Determine the convergence or divergence of the following series:

a) $\displaystyle\sum_{n=1}^{\infty} \frac{4}{3n^2 - 2n + 5}$

b) $\displaystyle\sum_{n=1}^{\infty} \frac{3^n}{5^n - 2}$

c) $\displaystyle\sum_{k=1}^{\infty} \frac{2\sqrt[3]{5k^4}}{(7k+1)^2}$

d) $\displaystyle\sum_{n=1}^{\infty} \frac{6n^8 + 5n^7 - 3n^2 + 2}{n^3(2n+5)^3(3n^3 - 2)}$

Ex. 3: Determine the convergence or divergence of the following series using any technique learned so far:

a) $\displaystyle\sum_{n=1}^{\infty} \frac{6n^2 + 17}{2n^5 - n^3 + 5n}$

b) $\displaystyle\sum_{k=0}^{\infty} \frac{2^k}{\pi^k + 7}$

c) $\displaystyle\sum_{n=1}^{\infty} \frac{n^2}{\sqrt{n} + 3}$

d) $\displaystyle\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 3}$

e) $\displaystyle\sum_{n=1}^{\infty} \frac{2n}{(n^2 + 5)^3}$

Which of the following series cannot be shown to converge using the limit comparison test with the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$?

(A) $\displaystyle\sum_{n=1}^{\infty} \frac{4}{3n^2 - n}$

(B) $\displaystyle\sum_{n=1}^{\infty} \frac{15}{\sqrt{n^4 + 5}}$

(C) $\displaystyle\sum_{n=1}^{\infty} \sin\left(\frac{1}{n^2}\right)$

(D) $\displaystyle\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$

AP Calculus II
Notes 9.5
Alternating Series

So far, the majority of the series we have dealt with have had positive terms. Now, we will look at series that have both positive and negative terms. This is known as an **alternating series**.

<u>Alternating Series Test</u>:

Let $a_n > 0$. The alternating series

$$\sum_{n=1}^{\infty} (-1)^n a_n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^{n+1} a_n \quad \text{or} \quad \sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

converge iff **both** of the following conditions are met:

1.

2.

<u>Demonstration</u>: Determine the convergence of the following using the Geometric and Alternating series.

$$\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n = \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{2}\right)^n =$$

$$\sum_{n=4}^{\infty} \left(-\frac{5}{4}\right)^{n-3} = \sum_{n=4}^{\infty} (-1)^{n-3} \left(\frac{5}{4}\right)^{n-3} =$$

Note that sometimes alternating series are hidden or appear to be alternating but are not. For example,

$$\sum_{n=1}^{\infty} (-1)^{2n} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^{2n+1} a_n \quad \_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_\_$$

while $\sum_{n=1}^{\infty} (-1)^{3n-1} a_n$ and $\sum_{n=1}^{\infty} \cos(\pi n)\, a_n$ ____________________________.

<u>THIS TEST CANNOT DETERMINE DIVERGENCE!!!!!</u>

<u>Ex. 1:</u> Determine the convergence or divergence of the following series:

a) $\displaystyle\sum_{n=1}^{\infty}(-1)^n \frac{11}{n+4}$

b) $-\frac{4}{1}+\frac{5}{2}-\frac{6}{6}+\frac{7}{24}-\frac{8}{120}+\frac{9}{720}-\ldots$

c) $\displaystyle\sum_{x=2}^{\infty}(-1)^{x+1} \frac{x+1}{\ln x}$

d) $\displaystyle\sum_{k=1}^{\infty}\frac{k}{(-3)^{k-1}}$

Absolute and Conditional Convergence

Conditional Convergence

If $\sum_{n=1}^{\infty}(-1)^n a_n$ converges but $\sum_{n=1}^{\infty}|a_n|$ diverges, then $\sum_{n=1}^{\infty}(-1)^n a_n$ is **conditionally convergent**.

Reason for Conditional: $\sum_{n=1}^{\infty}\frac{(-1)^n}{\sqrt[3]{n}}$

Absolute Convergence

If $\sum_{n=1}^{\infty}(-1)^n a_n$ converges and $\sum_{n=1}^{\infty}|a_n|$ converges, then $\sum_{n=1}^{\infty}(-1)^n a_n$ is **absolutely convergent**.

What is an alternating series that is absolutely convergent?

How could we use the Direct Comparison Test to further support the concept of absolute convergence?

Ex. 2: Determine the convergence or divergence of the following series. Classify any convergent series as absolutely or conditionally convergent.

a) $\displaystyle\sum_{n=1}^{\infty}(-1)^n\frac{n!}{2^n}$

b) $\displaystyle\sum_{n=0}^{\infty}\frac{\cos(n\pi)}{\sqrt{n+4}}$

c) $\displaystyle\sum_{n=1}^{\infty}(-1)^{n-1}\frac{\sqrt[3]{n}}{\sqrt{n}}$

d) $\displaystyle\sum_{n=2}^{\infty}\frac{(-1)^n}{n\ln n}$

e) Which of the following series is absolutely convergent?

(A) $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{2n}$

(B) $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{1}{\sqrt{n}}$

(C) $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\frac{n}{n+1}$

(D) $\displaystyle\sum_{n=1}^{\infty}(-1)^{n+1}\left(\frac{1}{2}\right)^n$

f) $\displaystyle\frac{5}{1}-\frac{6}{8}+\frac{7}{27}-\frac{8}{64}+\frac{9}{125}-\ldots\ldots$

AP Calculus II
Notes 9.6
Ratio Tests

This section begins with a test for absolute convergence known as the **Ratio Test**.

<u>Ratio Test</u>

Let $\sum_{n=1}^{\infty} a_n$ be a series with nonzero terms.

1. $\sum_{n=1}^{\infty} a_n$ converges absolutely if $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$.

2. $\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$ or $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \infty$.

3. The Ratio Test is inconclusive if $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1$.

<u>Connection to Geometric Series and Inconclusive Examples:</u>

The Ratio Test is not a test that works for everything, but it is very useful for series that converge rapidly. These include factorials and exponentials. It especially is useful when a series has multiple parts. It also does not matter whether the series alternates or does not.

Ex. 1: Determine the convergence and divergence of the following series:

a) $\sum_{n=1}^{\infty}(-1)^{n-1}n\left(\frac{10}{9}\right)^{n+1}$

b) $\sum_{k=1}^{\infty}(-1)^{2k+1}\frac{4^k}{k!}$

c) $\sum_{n=2}^{\infty}\frac{5n^2 2^{n+1}}{3^{2n}}$

d) $\sum_{n=1}^{\infty}\frac{(-1)^n(n+2)}{n(n+1)}$

Ex. 2:

BC3: Let $f(x) = -\frac{2}{\sqrt{x}}$ and let $A = \sum_{n=1}^{\infty} a_n$ where $a_n = f(n)$.

(**a**) Determine whether $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$ converge or diverge. Explain your reasoning.

(**b**) Consider $\sum_{n=1}^{\infty} b_n$ where $b_n = f'(n)$. Determine whether $\sum_{n=1}^{\infty} b_n$ converges or diverges.

Explain your reasoning.

(**c**) Consider $\sum_{n=1}^{\infty} h_n$ where $h_n = 2^{f(n)/f'(n)}$. Find $\sum_{n=1}^{\infty} h_n$ or show that it diverges.

(**d**) Consider $\sum_{n=1}^{\infty} k_n$ and $\sum_{n=1}^{\infty} m_n$ where $k_n = m_n a_n$. If $\sum_{n=1}^{\infty} k_n$ converges by the ratio test where

$\lim_{n \to \infty} \left| \frac{k_{n+1}}{k_n} \right| = \frac{2}{3}$, determine whether $\sum_{n=1}^{\infty} m_n$ converges or diverges. Explain your reasoning.

BC3: Let $f(x) = -\frac{2}{\sqrt{x}}$ and let $A = \sum_{n=1}^{\infty} a_n$ where $a_n = f(n)$.

(e) Consider $\sum_{n=1}^{\infty} c_n$ where $c_n = \frac{f'(n)}{f(n)}$. Determine whether $\sum_{n=1}^{\infty} (-1)^{n+1} c_n$ converges or diverges.

Explain your reasoning.

(f) Consider $\sum_{n=1}^{\infty} d_n$ where $d_n = \frac{f(f'(n))}{n}$. Determine whether $\sum_{n=1}^{\infty} d_n$ converges or diverges.

Explain your reasoning.

(g) Consider $\sum_{n=1}^{\infty} g_n$ where $g_n = f^{-1}(n)$. Determine whether $\sum_{n=1}^{\infty} g_n$ converges or diverges.

Explain your reasoning.

(h) Consider $\sum_{n=1}^{\infty} j_n$ where $j_n = f(n)f'(n)$. Determine whether $\sum_{n=1}^{\infty} j_n$ converges or diverges.

Explain your reasoning.

Ex. 3: Alrighty, summary time. Determine the convergence and divergence of the following series. If the series are alternating, determine if it is conditional or absolute. If you can determine what the series converges to, find the sum:

a) $\displaystyle\sum_{n=0}^{\infty} \frac{n+1}{3n+5}$

b) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{\pi}{6}\right)^{n+1}$

c) $\displaystyle\sum_{n=1}^{\infty} ne^{-n^2}$

d) $\displaystyle\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$

e) $\displaystyle\sum_{k=0}^{\infty} (-1)^k \frac{3}{4k+1}$

f) $\displaystyle\sum_{n=1}^{\infty} \frac{n!}{10^n}$

g) $\displaystyle\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{1+\sqrt{n}}$

h) $\displaystyle\sum_{m=1}^{\infty} (-1)^m \frac{2m^2-1}{1-m^2}$

i) $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^{n+1} e^n}{n!}$

j) $\displaystyle\sum_{n=1}^{\infty} \frac{5^n}{3+6^n}$

k) $\displaystyle\sum_{n=1}^{\infty} \frac{3n-1}{\sqrt[3]{n^5+6n+3}}$

l) $\displaystyle\sum_{n=1}^{\infty} \left(\frac{2}{n+1} - \frac{2}{n+3}\right)$